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1 Problems and Requirements of Chemical Protective Clothing

Scenarios where workers are exposed to hazardous materials require chemical protective clothing (CPC). Protection levels range from limited spray protection (Type 6) to fully encapsulating, gas-tight suits (Type 1) with self-contained breathing apparatus (SCBA) ["DGUV Information 212-019"].

Figure 1: Type 1 chemical protection suit 4005/NEO manufactured by isotemp[1]

Confined spaces are defined as areas in which, through their design or site conditions, "inadequate air replacement rates, or the substances [...] found within them or introduced into them, particular hazards exist or may arise that substantially exceed the normal prevailing hazard potential of workplaces" ["DGUV Regel 113-004"]. Tasks that require entering confined spaces are typically cleaning, inspection and/or maintenance of, for example, tanks in chemical plants, rail cars and highway tank trailers and even sewage containers. Traces of the contents may remain, even if their containers are prepared correctly. These residues might—despite being initially inactive—become dangerous through work procedures (e.g.

[1]Vollschutzanzug 4005/NEO · Chemie · isotemp

welding), agitation or chemical reactions [“DGUV Regel 113-004”], hence requiring the highest levels of protection. Other use cases are emergencies, such as confined space rescue, removing chemical spills or other accidents involving unknown, possibly hazardous substances [“OSHA Technical Manual Section VIII”].

These tasks are highly demanding—both physically and mentally—particularly because of the equipment. Since Type 1 suits are not gas-permeable, they compromise the body’s ability to regulate temperature, leading to exhaustion and heat fatigue even faster [Holmér, 2006]. The exertion is exacerbated by heavier equipment and heat stress, ultimately reducing work capacity [Smolander et al., 1984]. Maximum working time in a full-body suit with SCBA is therefore limited and might decrease even more depending on air and exhaustion levels.

Further, the material and shape of the suit muffle hearing and limit mobility and the field of view to such an extent that even trained personnel might need double the time to complete tasks when wearing CPC [Murray et al., 2011].

2 Augmented Reality Technology

Augmented reality (AR) is a subcategory of extended reality (XR), next to mixed reality (MR) and virtual reality (VR). In AR virtual objects are "superimposed upon or composited with the real world" [Azuma, 1997], in opposition to virtual reality which—by replacing the real environment entirely—creates the deepest sense of immersion. Mixed reality combines features of both, the main distinction being the ability to interact with virtual objects in the real world [Hari Chandana et al., 2023].

Vi et al., 2019 Speicher et al., 2019

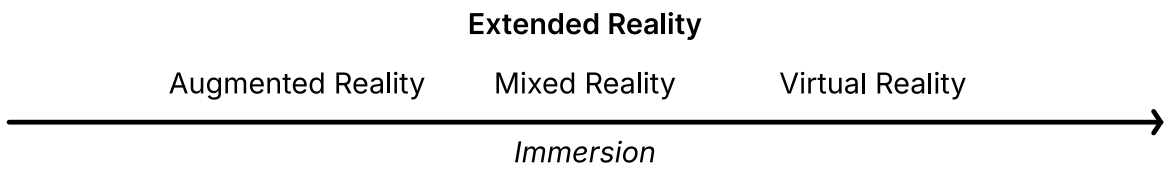

Figure 2: The extended reality spectrum, encompassing augmented, mixed and virtual reality.

Azuma (1997) defines three requirements for applications to be classified as AR. They combine real and virtual content, they are interactive in real-time, and the virtual objects are registered in 3D space. With the technological advancements since the 2000s these requirements are understood more as guidelines than rules.

A head-up display overlays information onto reality in the user's line of sight to avoid the need to look away from their primary task to acquire needed information. Originally used in military aircraft, today HUDs are built into consumer vehicles as well. Apart from these integrated HUDs, content can be presented on a head-mounted display. This affixes the display to the head, typically as a helmet or glasses, enabling mobile usage. Contrary to the above definition by Azuma (1997), the content displayed on a HUD is two-dimensional and head-stabilized, rather than anchored spatially, i.e. world-stabilized (see figure 3).

Manufacturers of AR and VR headsets recommend avoiding classic 2D interfaces, because it breaks immersion. The planned use cases for this HUD require the opposite—to be able to focus on the primary task in the real world.

The most important feature of head-stabilized content is that it is consistently found in the same location. World-stabilized elements are helpful for exploring and better for immersion. When the needed information is unrelated to the environment, head-stabilized display might be better. Another reason for head-stabilized display

is that it is hard to orient information when moving around.

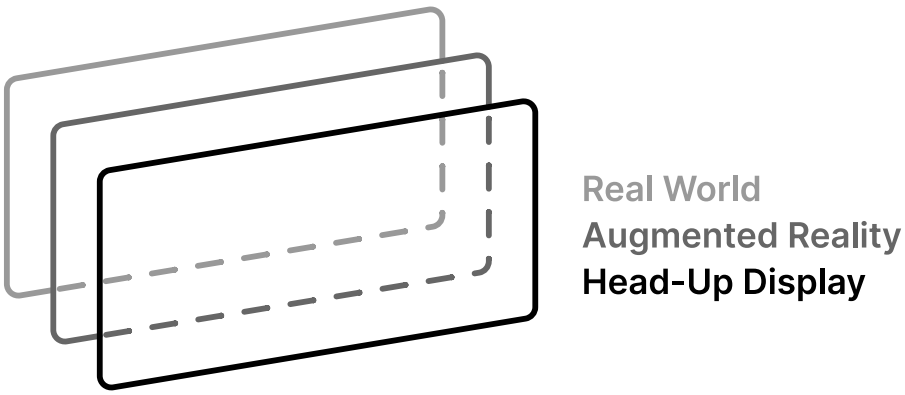

Figure 3: Visualization layers of head-mounted displays: reality, Augmented Reality (3D, world-stabilized) and User Interface (2D, head-stabilized).

A head-mounted display can cover one (monocular) or both eyes. Depending on the technology, both eyes receive the same image (biocular) or each will be shown a different image (binocular). In a monocular system, both eyes receive different information, since the HMD is placed in front of only one eye. This introduces the problem of binocular rivalry. With a biocular display, the same image is presented to both eyes, which affects depth perception. Binocular presents a different image to both eyes CAKMAKCI 2006. The depth perception here is affected by the vergence-accomodation conflict. also causes eye strain.

2.1 Technical Challenges Affecting User Interface Design

Limitations unique to optical see-through displays will be discussed in the following section.

Field of View. Syberfeldt et al. (2017) puts the boundary for an acceptable field of view at 30° at minimum in industrial use cases, which current systems can fulfill. Microsoft's HoloLens sits at 43° by 29°, a few systems hit around 40° (https://vr-compare.com/ar).

Screen Shake. As with all head-worn devices, HMDs are suceptible to shaking when walking. Janaka et al. (2023) found that this movement causes legibility issues. This problem can be avoided by using fonts with thicker strokes (both horizontal and vertical) [Matsuura et al., 2019].

2.2 Optical See-Through Displays

There are two approaches for combining virtual content with reality. With video see-through, information will be superimposed on a video stream filmed by a camera on the headset. These headsets fully occlude sight, the stream replaces the outside view. An optical see-through system allows the user to directly view reality. Here, optical combiners

There is no weighing of advantages here. When a video see-through system fails, it will be impossible to see. Failing equipment must not impair vision, especially since it cannot be taken off in a fully encapsulated suit.

2.2.1 Ambient Light and Contrast Issues

Regardless of optic technology, images are formed by combining light, meaning the display competes with ambient lighting for visibility, which is especially problematic in outdoor conditions. Most established OSTs provide around X lux, compared to overcast winter sky ca. 6000 lux, sunlight > 100.000 lux. The biggest issue for legibility is missing contrast in bright environments. Especially outdoors, visibility of current OST technology is poor. Kerr et al. (2011), for example, report that participants repeatedly used their hands to shield the display from external light to be able to read the content on the display.

This follows the additive color model, where adding all wavelengths of light together results in white. Black is defined throught the absence of light. With physical screens, the color we see is the actual color of the display material. Since OSTs have no physical screen, "black" is transparent [Gabbard et al., 2013]. This also means that darker colors are unavailable for designing on OSTs, since they are mostly translucent. Many current OSTs like [MODEL] have shaded lenses to improve the contrast somewhat Itoh et al., 2021. Unfortunately, current AR technology is not capable of rendering the contrast levels necessary for outdoor use [Erickson et al., 2020]. Technical solutions in development include OSTs that can partially occlude areas of the display to block light. Still very experimental [Davari et al., 2020].

2.2.2 Color Blending

Because of the display transparency, the background impacts colors [Itoh et al., 2021].

2.3 Methods of Interaction

Due to not being constrained to two dimensions, XR applications can explore new types of interaction, such as gesture recognition, speech recognition and gaze tracking.

Krevelen and Poelman, 2010 WIMP only for 2D, AR needs six degrees of freedom (translation and rotation XYZ)

Hari Chandana et al. (2023) emphasize the importance interactions being intuitive and feeling natural.

Industrial use cases require hands-free interaction methods—i.e. speech and gaze. Due to the often very noisy environments in industrial environments, gaze tracking is more reliable.

3 The Human Visual System and Attention

3.1 The Human Visual System

A simplified model puts the field of view to around 200° to 220° horizontally and around 125° vertically with the vertical field extending slightly more downwards [Loschky et al., 2017]. The central vision covers about 5° around the point of fixation, where acuity is highest [Loschky et al., 2017, Rosenholtz, 2016]. The remaining field is the periphery, with boundaries at 30° and 60° dividing it into near, mid and far peripheral vision.

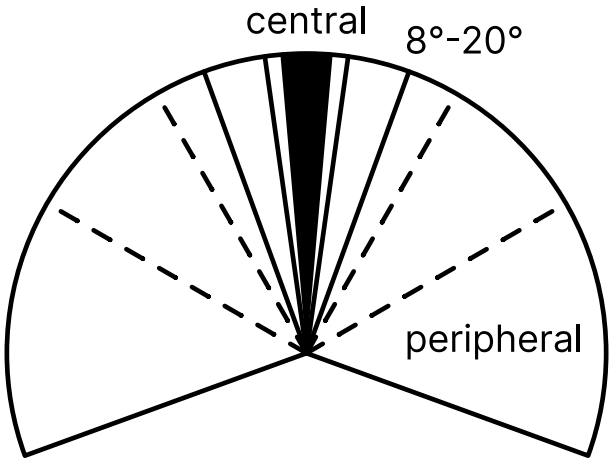

Figure 4: Simplified diagram of the field of view.

Despite changing anatomical structure from center of gaze to periphery, entire scene appears in high resolution, contrast, color saturation Loschky et al., 2017.

Perception does not require a lot of details, the recognition of abjects and orienting oneself ist still possible [Rosenholtz, 2016]. That is why image thumbnails are still recognizable.

Rosenholtz (2016) suggests that crowding is the biggest issue when it comes to parsing information. The effect is more intense if the target and its neighboring elements are similar (e.g. the same color and shape) [Pelli and Tillman, 2008]. The farther away elements are from the center of vision, the bigger the distance between elements has to be.

The peripheral vision is good at detecting motion, provides better sight in low light conditions [Johnson, 2014] and contributes to functions like orientation, navigation, balance and hand-eye coordination. Restricting the peripheral vision affects all of

these negatively, getting worse the more is occluded, severely impacting performance Alfano and Michel, 1990.

Color sensitivity decreases toward the periphery, red-green moreso than blue-yellow colors and sensitivity to luminance. Hansen et al. (2009) found that as long as the stimulus is of sufficient size, perceiving color is possible even at 50° eccentricity. Most modern OST-HMDs feature a 40° field of view, meaning the full available color spectrum can be used.

3.2 Visual Attention

The perception of images generates too big an amount of information to process at once. This bottle neck is solved by visual attention: limiting the processing by focusing on just a fragment of the current visual stimuli [Anderson et al., 2005, Rosenholtz, 2016]. This process is guided by the peripheral vision, subconsciously preselecting salient features of the environment [**<empty citation>**]. Saliency describes features which make a target differ from surrounding objects [Johnson, 2014].

Rosenholtz, 2016 - "Because of the brain's limited capacity, our visual systems cannot automatically perform every possible visual task at the same time. Instead, the brain may concentrate resources on one task in a given moment, then switch to focus on another task." - "Our results suggest that the visual system looks for evidence of a change throughout the visual field, possibly in parallel, rather than looking for a change merely where one fixates. However, that evidence of a change can be weak because of loss of information in peripheral vision, making change detection difficult."

Cater et al., 2002 - process by which we humans select a portion of the available visual information for localization, identification, and understanding of objects in an environment - allows the visual system to process visual input preferentially by shifting attention about an image, giving more attention to salient locations and less attention to unimportant regions

There are two ways in which an object can be focused: bottom-up and top-down. Bottom-up attention is subconscious, stimulus driven. For the top-down process, the brain directs the eyes to focus on a consciously chosen object. Interestingly, when concentrating on an objective, the top-down attention lessens the effect of distracting stimuli [Connor et al., 2004].

If multiple stimuli compete for attention, even salient targets might be ignored, if they are irrelevant to the current objective. This is called inattentional blindness [Jensen et al., 2011]. This effect can be leveraged, for example Cater et al. (2002) who use it to only render high quality images in the center of vision.

4 Data Visualization

Visualizating data transforms it into information. Data itself is just noise, whereas information holds value. In what way data needs to be transformed is determined by context [Tergan and Keller, 2005].

The focus of data visualization is to make large datasets understandable. Most fundamentals explain how to chart this data well. Interpreting a chart needs time and cognitive space that is not available in scenarios like the confined space entry described in section 1. In these situations a clear picture of the current status has to be available immediately. Expanding general design foundations with some aspects of data visualization can still improve the quality of such status displays.

4.1 Applicable Concepts of Data Visualization

Visual variables correspond to the fundamental design elements bertin: position, size, shape, color hue, color value, orientation, texure

nominal, ordinal, quantitative data

Information Encoding

+ The type of visualization depends on the purpose of the information. Depending on the character of the data, different methods of visualization could be chosen.

Comparable to basic dashboard data, only the value of the parameter is relevant, since there is no relationship between the data. At most it is the relation between a current and a past value of a parameter.

Purposes of color to cenvey information are grouping or relationships. Grouping is necessary for the HUD, but is managed by the spatial location of and the spacing between elements.

Wilke, 2019

Since the presented scenarios only allow for single data values, the amount of possible visualization types is minimal. Ferdio, an agency specialized on infographics and data visualization, has created a library of visualization types[2]. In this library, out of a 166 options, nine are for single data.

[2] Data Viz Project | Collection of data visualizations to get inspired and find the right type

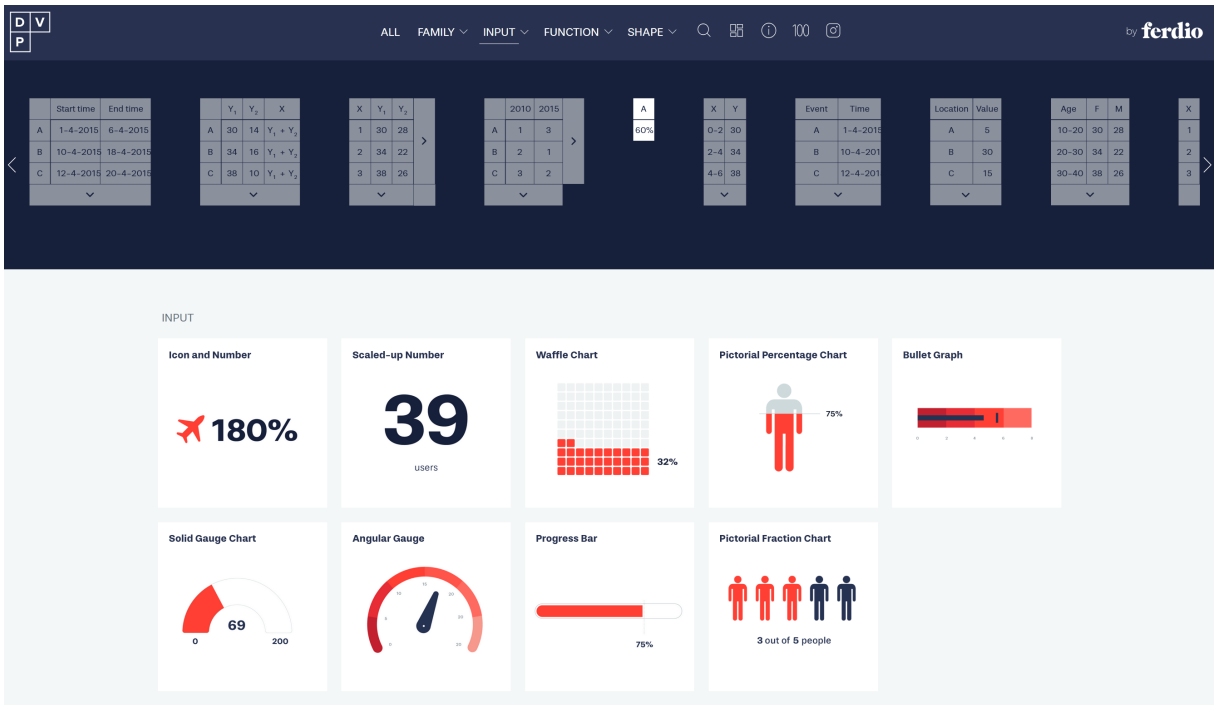

Figure 5: The options for single data (highlighted in the menu bar) in Ferdio's data visualization library.

4.2 Situation Awareness

Making good judgements becomes increasingly difficult when performing demanding procedures while being under stress and maintaining awareness of the surrounding area. Endsley (1995) highlights the role of situation awareness as a foundation for good decision-making in dynamic situations. She outlines three levels of SA, namely perception, comprehension and projection, of which perception is the most significant for UI design. This stage requires collecting the information relevant to the situation. In level 2 this information will be analyzed to form "a holistic picture of the environment, comprehending the significance of objects and events." Finally, likely outcomes can be determined by understanding the dynamics of the situation.

It is evident that situation awareness is therefore dependent on the processes of attention described in section 3. Subsequently, this illustrates the necessity of a thorough understanding of the pieces required for achieving good situation awareness to design a well-functioning HUD. The amount of information that can be rapidly and accurately obtained is directly influenced by how it is presented. In addition to prioritizing and organizing that information around task objectives, Endsley (1995) mentions the advantage of systems providing processed information rather than the data it is based on (e.g. displaying remaining time based on current

oxygen amount and usage).

Special attention should be paid to reserve highly salient features to critical events that truly necessitate changing the course of action. Further, it is vital to maintain a balance between showing all relevant information, while simultaneously avoiding visual overload. While looking for inaccessible information, other important content may be overlooked. Conversely, "filtering of extraneous information (not related to SA needs) and reduction of data (by processing and integrating low-level data to arrive at SA requirements)" is necessary to prevent clutter. This reiterates the importance of accurately evaluating and prioritizing task-relevant information.

Considering these recommendations supports evaluating the interaction of all components, since emphasizing some elements inevitably comes at the expense of others. Assessing the design concept by the amount of situation awareness it provides serves as the last step of ensuring the effectiveness of the system.

4.3 The Concept of Glanceability

It could also add the capability to summon more information when looking at an anchor. This method was tested by Lu et al. (2020).

context awareness Krevelen and Poelman, 2010 Context-Aware interfaces display information based on current task or cognitive load Rhodes, 1998. The risk of false positives is unjustifiable in safety-critical environments.

Instead of showing the UI at all times, the elements could be positioned outside of the HMD's field of view, to be accessed by either looking or turning the head in the direction. Lu et al. (2020) explored and tested this concept, naming it "Glanceable AR".

- related to gaze-based interaction - head-turning method requires body position tracker to enable sensing the head turning in relation to the body

Would be helpful for secondary information [DEFINE secondary info]. To avoid the user needing to remember in which direction the information was placed, this information could be anchored by representative icons. Or the primary information which hides related information (e.g. task list that shows current and a limited amount of next steps, that displays the complete list when accessed)

LU 2020, 2023 - "designing glanceable interfaces requires careful consideration

not only about how to present information visually, but also about how to access the information effectively" Lu et al. (2020) proposes four design principles: "DP1. Virtual content should be fixed to the user. DP2. Virtual content should be spatially distributed in the periphery. DP3. The interaction to access virtual content should be hands-free. DP4. Information should be accessed by glancing quickly at the periphery." - voice input: disturbs others in public, HERE: radio comms, loud industrial env - "we define "periphery" as the space at the edges of or beyond the field of view (FoV) of the AR display" - eye-glance interface (HUD) - baseline, does not require sensors or trackers - "benefits of information being highly visible and accessible, but it may also be obtrusive, occlude the real world, and cause the problem of information overload" - always-there information might interfere with peripheral awareness - head-glance - content outside FOV, fixed to body, turn head to access → clear view - requires to look away from forward FOV momentarily, impacted performance of primary task in study (keeping a specified distance to virtual human when walking) - HYPO might be problematic when moving quickly (emergency) - mental demand higher in study, possibly because of - frequency of information access, leading to a lot of extra head movement avoidable by other methods - need to remember location of information - gaze-summon (look at indicator for 0.5 sec to display information) - additional time needed to access information results in unusability for industrial use case - possible eye strain from keeping gaze on target - increase of false positives when participants took a turn (reflex to look into intended direction + eye-tracking jitter from quick head movements) Lu et al. (2020) mentions the display being less impactful when participants concentrated on the primary task. CONCENTRATION LEVEL? - CON primary task of walking not comparable to task with high cognitive load + physical demands

Peripheral Displays - there's often alternative forms of information display involved, especially hardware design more resembling art objects than traditional computing devices - ubiquitous computing: computing everywhere in any format

Although peripheral displays are usually mentioned in tandem with ubiquitous computing, relating concepts are applicable here. The first is context awareness of systems, The second, and arguably more relevant, is attention management.

MATTHEWS 2006 - peripheral/ambient displays "move information from the periphery to the center of human attention and back" - "Peripheral displays receive varying amounts of attention depending on how easily users can interpret the information they convey." - this seems interesting - does this mean you pay less attention when the information is easier to parse? - "Glanceability is critical to peripheral display design because users need to quickly glance at and read dis-

played information with minimal interruption to their primary task." - MATTHEWS 2005 distinguishes between peripheral and notification displays, which are used at the operation and action level respectively. - this concept includes both as "display modes" - "Design variables used for capturing user attention include abrupt onset, flashing, bouncing, and other motion." - "Glanceability is greatly affected by the amount and type of information being displayed." - visual search processes bottom-up (raw sensory input) vs top-down (goal-driven) CONNOR 2004 → that's the vocab for the passive/active modes i had in mind! - "Visuals that support top-down search will be easy to cognitively interpret. Visuals that support bottom-up search enable relevant information to pop-out. Baldassi and Burr, 2004" - "Pop-out is a bottom-up drawing of attention to an object, which occurs when an object within the visual field is distinctive along some visual dimension (for example, possessing a distinctive color or brightness when compared with other objects in the field)." → POWER OF CONTRAST - "If visual features are to be used successfully in glanceable displays, they will need to balance distinctiveness with too much demand for attention." - variables were used to distinguish between versions of otherwise the same items (e.g. email categories of notifications) - task description after Chewar et al. (2004): notification, reaction, comprehension - HYPO icon position for notification dependent on display placement direction (i.e. icons on right side when display is on the left) - information types: ordered, unordered finite, unordered infinite - principles for representation - semantically meaningful - abstract - distinct

Ishiguro and Rekimoto (2011) presented a method to show a simple icon that transitions to related information when gazed at.

Birnholtz et al., 2010 peripheral displays

Notification levels equal importance of information, corresponding to action priority (in HUD case) TABLE notification level and corresponding design choices + elements e.g. change blind, make aware, interrupt, demand action x size, location, color, motion (+ frequency),...

- demand action requires user to perform an action to silence the alert → will not be used in HUD - could argue that oxygen low or hazard proximity demands action, but in in emergency cases there might be no way to immediately take action based on that, and the HUD should not impede the user's decision-making, which an ongoing alarm would do

passive/ambient/top-down By default, the display just shows necessary information. This mode is designed to not get the user's attention so they can focus

on their primary task. Another reason to avoid color and movement here is to have the maximum contrast possible if the user needs to get notified of something.

Janaka et al., 2023 getting attention without harshly interrupting primary task Salisbury et al., 2024

interruption/active/notification/bottom-up/ Certain functionality to get the user's attention is implemented. warning pops up in FFOV [Rosenholtz (2016)] + then moves to the side (to not hinder visibility) w/ blinking icon (so it's not forgotten)

5 Design Guidelines

Following scenario requirements, technical limitations and human factors, the HUD needs to fulfill the following demands:

Ideally, the finished system shows as much information as possible/necessary without limiting situation awareness, causing fatigue or sensory overload. The displayed information should be comprehensible in as short a time as possible.

In an emergency, the system must be able to get the user's attention, without hindering decision-making or interrupting the primary task completely.

Referencing rules for visual hierarchy, the usage of contrasts heightens difference between the two modes by limiting color and movement to emergency functions to get and keep attention without hindering reaction times.

The following design guidelines are tailored to ensure the best visibility and information acquisition possible.

Industrial equipment, especially if it might be easily damaged, technology for augmented reality gets more expensive with either bigger field of view or better color rendering. Making sure the proposed design foundations work in monochrome as well enables choosing more economic options, aside from accessibility.

Vicente and Rasmussen, 1990 **novice vs expert users** users are/will be trained with equipment → able to forego certain design principles is there a distinction between end user and professional/trained user? shneiderman (user interfaces): novice + expert user, universal usability HYPO acceptance of HUD is better with newly trained people regarding mental load, since they don't need to change existing patterns

5.1 Color and Contrast

The use of color is mainly limited by current optical see-through technology. As mentioned by GABBARD, colors with high source luminance will be visible best. Tying into economic reasons, designing the HUD functions to work with one color only ensures affordability as well.

Limiting the UI to one color makes sure the design works in all aspects (e.g. using

motion to get attention before color). Another reason is the economic factor: especially in scenarios where it's likely that equipment needs to be replaced because of damage or contamination, the acquisition cost should be as low as possible. One way to achieve this is to only have a monochrome display. So, the HUD UI should be able to function without accent colors. This also avoids impairing users with color blindness.

The library chroma.js can sort colors by luminance in the RGB color space, based on the Web Content Accessibility Guidelines 2.0. Sorting the base colors red (#FF0000), green (#00FF00), blue (#0000FF) and their intermediaries cyan (#00FFFF), magenta (#FF00FF) and yellow (#FFFF00) puts them into the following order (darkest to brightest): blue, red, magenta, green, cyan and yellow.

Figure 6: Base colors sorted by luminance.

https://gka.github.io/chroma.js/#color-luminance

https://www.w3.org/TR/2008/REC-WCAG20-20081211/#relativeluminancedef

https://rapidtoolset.com/en/tool/colors-sorter

These results can be confirmed via the OKLCH 3D model as well: input constant, high lightness and any hue, chroma peaks are yellow, green, cyan. Full chroma is needed to avoid colors blending together.

https://oklch.com/

The following design will use white, since it is the best visible color on an optical see-though display. If color is applicable, yellow will be used as accent color, mainly reserved for notifications, to heighten the contrast with the standard mode. Following Gabbard's study, red seems to be suboptimal, since it is strongly susceptible to color shifting.

5.2 Iconography

Needed information is found quicker when presented as graphical content.

- amount of coded information

Icons can utilize small screen estate more efficently.

- plane-based icons - another reason goes back to MATSUURA's typography study: lines are gonna be harder to read

LIN 2016 "Well-designed graphical icons can achieve such positive effects as drawing users' attention, shortening search time, reducing operational errors, and relieving users' burden." - plane-based icons are recognized easier - line-based icons dissect the space more often = more complex - complexity lowers recognition performance → even more if backgrounds adds another complex layer - white on black icon performed best

JANAKA 2023 - icon-augmented notifcations - significantly higher primary task performance than text - reduced interruption while maintaining reaction and comprehension - when the primary task's complexity increased or needed higher attentional control, users familiar with icons could understand pictogram format better than text format. - In such specialized domains, it is easier to defne a set of well-designed icons that become highly familiar to users through repeated use - icon-augmented notifcations are easier to perceive in suboptimal conditions during mobile OHMD usage. Specifcally, even though virtual content on OHMD is blurred when users focus on the physical world, i.e., limited OHMD focal distance [47, 48], the pictogram format on OHMD can be easier to recognize than the text format - higher salience during shakes, be easier to perceive in sub-optimal conditions (sec 9.2.3), and provide higher attention control

Figure 7: Line-based (left) and plane-based (right) icons.

Especially due to the need of thicker strokes, an icon's readability might be impacted. Simpler icons with varied silhouettes offset this.

Figure 8: Adjustment for the readability of the explosive hazard icon.

5.3 Typography and Stroke Width

Aside from screen shake, reading performance is further impacted by lighting and background texture. To find requirements for font size, RENKEWITZ 2008 tested the effect of "background texture and font color on text recognition". They set a benchmark of two seconds as threshold for fast recognition, which resulted in font size thresholds from 23 to 42 pt, depending on color and background. These numbers might have to be adjusted for an OST-HMD, since they used a video see-through system, and red and blue as colors, both colors with lower luminance.

The typeface "Inter" fulfills all of the requirements. It has a uniform thickness of strokes, square letter width and good spacing between its lines.

Naturally, the same requirements exist for the stroke width.

Figure 9: A comparison of different stroke widths.

5.4 Layout and Spacing

As mentioned in PERIPHERAL VISION, crowding affects information acquisition the most. It seems advisable to prefer wider spacing between elements. limiting the number of elements per side to 3 so they can be spatially anchored to the corners and one slot in between.

Chua et al., 2016 - ! monocular OST-HMD - middle + bottom center positions were noticed quicker - top + peripheral positions were more comfortable, unobtrusive - center-right is mentioned as best balance between usability + comfort

Katsuyama et al., 1989 - studied task requiring frequent eye shifts - (1) displays located 15° below a primary viewing area are better perceived than identical displays located 15° above - (2) ratings of discomfort involved in eye movements (necessitated by the viewing of physically separated displays) correspond to their performances on the secondary task - task performance decreases the further the distance from center

Kim et al., 2019 - tasks that require referencing information - "performance, workload, and usability were more affected by UI designs than the HWD type." - "Compared to the baseline paper pick list condition, using an HWD, especially with the graphic-based UI, led to improved job performance and reduced perceived work-

load, while causing increased discomfort" - "Responses to the workplace cognitive failure scale also indicated a lower risk of cognitive failure specifically with the graphic-based UI design + always-on display."

DYSON 2001

According to Lin et al. (2021), 8° to 20° horizontally seem to be most comfortable. Vertically limit to 8° above and below visual center - "We are surprised to discover that there are no significant differences in task times between the four different display positions. We expected a more uncomfortable viewing angle would have a more dramatic effect in slowing users so that their order picking times would increase, especially for the bottom-right position." BUT "we realized that users stop after picking one book to find and plan their path to pick their next book, instead of trying to read and interpret data while on the move. The extra time taken to stop and plan their path could mask the differences in time taken to perceive picking information" - "the bottom-center position could be inferior in situations in which users have to look at their feet such as stairs mobility and rough terrain maneuvering." - "A bottom-right display could also interfere with tasks involving the hands." - "The display situated between 5.4° to 14.6° toward the ear was most preferred by users. It did not block the line-of-sight view and was also closest to center-center positioning. However, Dowell's work on cognitive capture [8] suggests keeping all pixels at least 8° from line-of-sight." - "there may be an ideal, general purpose HWD position between 8 to 20° temporally offset from line-of-sight. Horizontal FOV would thus be 12°. Vertical FOV requires more investigation, but limiting pixels to between ±8° from center as in Haynes's comfort study [21] seems wise" - "As mentioned previously, from Chua's work we wish to avoid nasally offset displays, and positions much above line-of-sight are uncomfortable for extended use CHUA 2015." - used a monocular display? with 12° x 16° FOV, theorize that they could put another display on the other side, doubling the FOV, and users can switch between → results should be applicable to HUD

Dowell et al., 2002 - flight sim - "HUD information located less than 8 deg from the out-the-window point of interest can induce cognitive tunneling and impair performance on the task over which the HUD is superimposed" - "Additionally, it was shown that HUD symbology placed in the upper portion of the visual scene (greater than 8 deg from the central focus), led to improved simultaneous processing of the HUD symbology and the forward view." BUT "hypothesized that this difference is rooted in the reduced clutter and background motion found when the HUD overlaid the sky, as opposed to the ground."

5.5 Notifications

The billboard style occludes much of reality around the text, which should generally be avoided. However, exactly the feature makes it suitable for high priority notifications.

Matthews et al. (2004) mention three ascpects of attention management in their "Peripheral Displays Toolkit": abstracting information to make it glanceable (see GLANCEABILITY concept), assigning notification levels to information and designing (visual) and transitions between those levels.

Screen edge indicators might not work.

—

Abstraction of Information

optimal information density: shows all relevant information without over whelming the user

GRAPHIC abstraction as scale of readability ↔ unobtrusiveness

Information can be abstracted when the represented concept is simpler (e.g. ambient motion detector), in which case primitive elements of design, such as color and movement can be sufficient.

5.6 Technical

Eye tracking could be used to dim the HUD while the user does not look at it. This would decrease distractability, and reduce glare in darker environments.

6 Conceptualization

6.1 Relevant Parameters

A prioritization of elements is necessary for decisions on visual hierarchy, as well as which elements can be left out if there is not enough space.

	Display Mode	Visualization
Oxygen	Continuous	Status Bar, Value
Hazard Proximity	Pop-Up	Icon
Warning Notification	Pop-Up	Billboard
Working Time	Pop-Up, Continuous	Countdown
Heart Rate	Continuous	Icon, Value, Trend
Ambient Temperature	Continuous	Icon, Value, Trend
Humidity	Continuous	Icon, Value
Checklist	Continuous	Line Highlight
Tool Help	Pop-Up	Icon, Value

Table 1: Parameters chosen for visualization, ordered from most to least important.

maybe columns: frequency of checking,

Continuous means that the component will be displayed permanently. Parameters marked with pop-up will be displayed occasionally when relevant.

6.2 Components

Following the previous guidelines, every element is designed for maximum clarity.

6.2.1 Visualizing Oxygen Usage Through a Status Bar

Since the oxygen value is the crucial factor whether or not to continue work, this is the component taking the spotlight. As the most critical parameter, this component will be visible in any version of the HUD. To check the current status as quickly as possible, the value will be visualized with a status bar.

Visualization helps to avoid mental math.

graphical vs text: status bar is understood faster than the number amount + unit

The first decision was the way to visualize the status bar. Mechanical pressure gauges with mechanical pointer show visual progress by default. According to Xia et al. (2025), bar charts outperform radial visualization.

Figure 10: The base component with configurable icon, parameter value, unit and trend indicator.

The second version implements the status bar in four segments, making it easier to estimate the amount left. Another hypothetical advantage of the gaps in the bar might be that more of the background can be seen.

Figure 11: The base component with configurable icon, parameter value, unit and trend indicator.

Depending on screen real estate, there are horizontal and vertical versions of the component.

In the horizontal version the label is placed in front of the bar so as not to have another "reading bump".

In the vertical orientation, the label and value are aligned at the top and bottom respectively, to account for visual crowding.

The oxygen value is located so that the status bar is seen first, e.g. if the component is on the left, the bar will be on the right side. This helps to maintain an optical guide.

- time left w/ current breathing rate

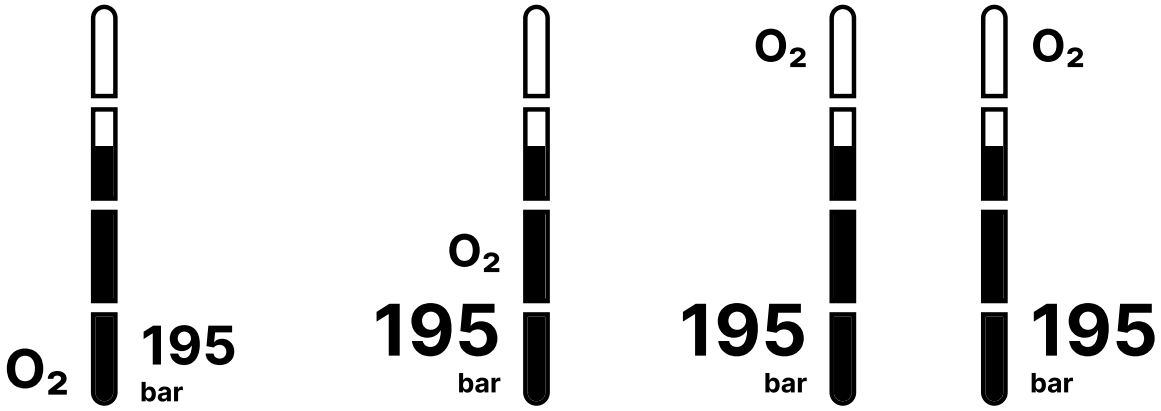

Figure 12: The base component with configurable icon, parameter value, unit and trend indicator.

6.2.2 The Base Component

This element is built with a simple configuration for secondary information that is helpful but does not need to be prioritized. This includes factors like heart rate, but also environmental information such as ambient temperature or humidity.

Values that are not to be visualized are similar enough to be based on a universal base component. Apart from the actual parameter value, any of these elements can be hidden. This allows flexible configuration depending on the type of information.

The icons replace labels, and should therfore be as unambiguous as possible. If the icon is unambiguous enough, it is easier to hide the unit. In some cases, the unit supports the identification of the value. Leaving the unit out saves space as well.

The trend indicators are displayed based on the difference of the current and previous value. If the value rises or falls an unusual amount, which could note environmental events. For dynamic elements like these, the frequency of changes should not be too high to avoid distraction.

Comparing the short and long arrows, the long arrows are better readable when out of focus.

Figure 13: The base component with configurable icon, parameter value, unit and trend indicator.

Figure 14: Example configuration of the base component. Left to right: heart rate, time and tool help.

The time component is a special case: it will only be displayed after a configured amount of time. The element popping up is meant to be perceivable in the peripheral vision, marking (for example) the halfway point. This action alone is therefore meaningful. Afterwards it can be checked consciously.

Since working time is limited by various parameters and therefore not fixed, it is a countdown. To avoid pulling unnecessary attention, the display is limited to minutes, instead of ticking down by the second.

6.2.3 Notifications and Pop-Ups

Instead of unique silhouettes, the hazard warnings are triangular to be more intuitive. The placement is therefore more important for peripheral distinction. Both icons are placed close together so that they are visible at the same angle if multiple hazards are present. Similar to a car's dashboard warning lights, each of them has a fixed position, facilitating quicker information acquisition. With regular notifications—depending on which appeared first—the order of content would be different.

Horizontal versus vertical arrangement Since the visual search field is wider than it is tall, horizontal might blend together, stacked vertically—therefore at the same visual angle—they might be more distinctive. More a loose guideline, flexible in regard to screen real estate.

Figure 15: Example notification with billboard style.

The core example of the notification system is the high priority warning. This component is designed with a so-called billboard style, meaning text with a surrounding box. The big, bright area ensures high visibility even if the pop-up is not

in the center of vision. Since it is the highest priority, it is the only element placed in the upper part of the screen. To avoid blocking central vision for too long, the pop-up will transform into a smaller version at the upper edge of the screen.

Grabs attention first, readability second step → text case? Oxygen vs O2/abbreviations better?

6.2.4 The Checklist Component

Since this element is text-based, it relies less on peripheral vision. To anchor the gaze the current task is centered on the screen and more distinctive than the other displayed tasks. Though the other tasks are smaller, everything is left-aligned to—once again—create an optical guide.

sentence case vs lower case

Note that this component is meant as a visual example, which is why certain aspects of functionality might not work in regard to handling the list. (e.g. if only two next tasks are displayed, there is no way to see if new tasks got added → task number display in dynamic situations, generally better for protocol)

The icon to the right is an anchor for possible gaze-based interaction, in this case to show more details. This design keeps the current task centered in the detailed view as well for faster orientation. This also mitigates side effects from possible false positives—i.e. if the system mistakenly opens the detailed view.

6.3 UI Examples

Note: the following illustrations are not to scale.

The maximum variant of the interface was designed with full functionality in mind, based on Microsoft's HoloLens 2[3]. This includes a horizontal field of view of 43°, a resolution of 1440 by 936 pixel per eye and full color availability (within the technical limits mentioned in section 2).

The layout is based on a three by three grid, anchoring components to the sides and corners, leaving the center free.

[3]

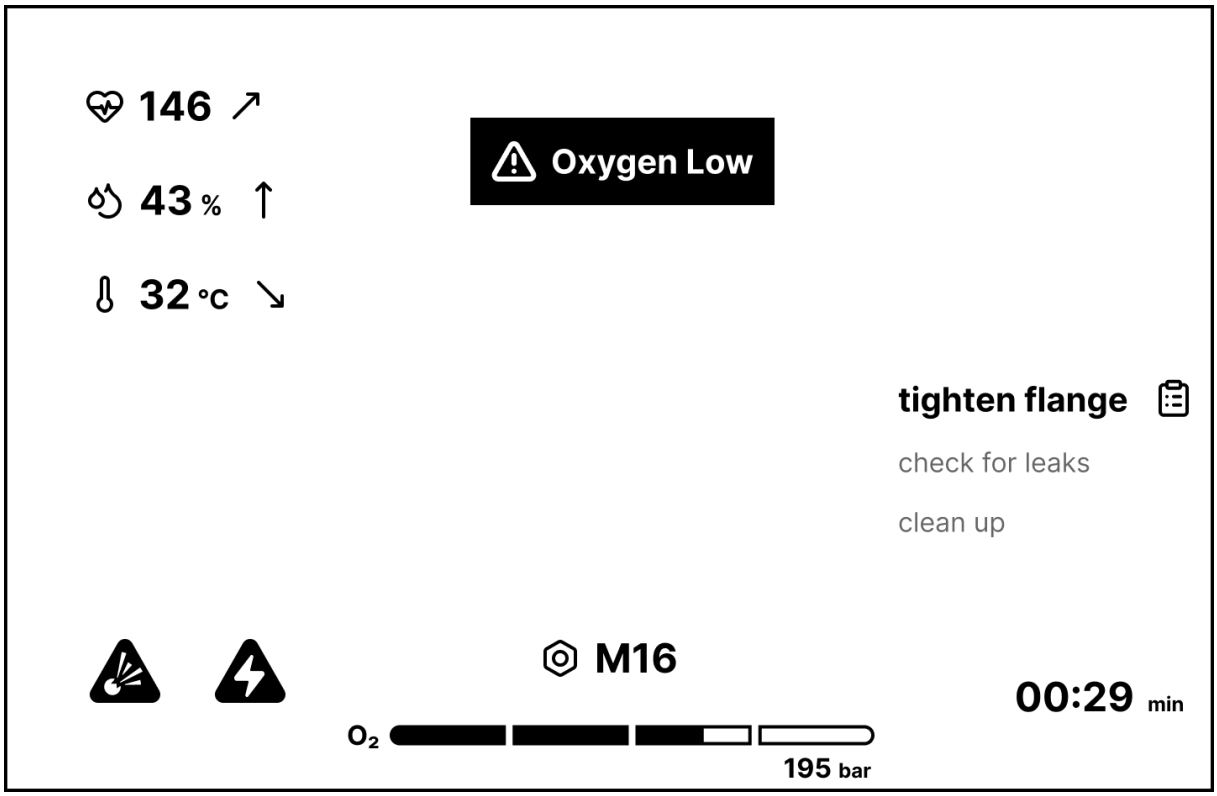

Figure 16: Full-featured UI concept for a binocular HUD.

While oxygen is the most important element, it is not necessarily the one that is checked the most, but it should be checkable the quickest. → lower center seems good KATSUYAMA 1989

Occasional information is placed above the oxygen bar, for the same reason: looking down is faster than looking up.

The minimum version is based on the Google Glass, featuring a monocular display with a resolution of 640 by 360 pixels.

https://vr-compare.com/compare?h1=EkSDYv0cW&h2=EnqtMtA05

A point could be made for overlapping components somewhat to use the space better. However, this would disrupt the grid pattern and optical guides, resulting in longer search times.

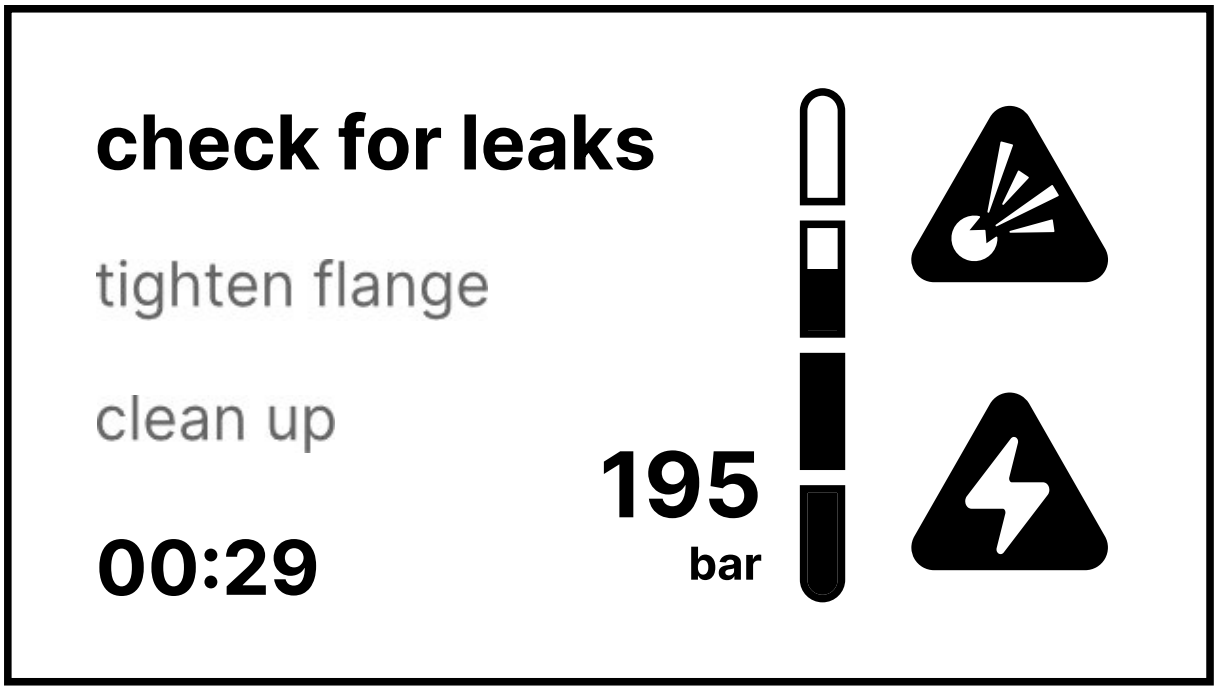

Figure 17: UI concept for a monocular HUD, worn in the upper left of the field of vision.

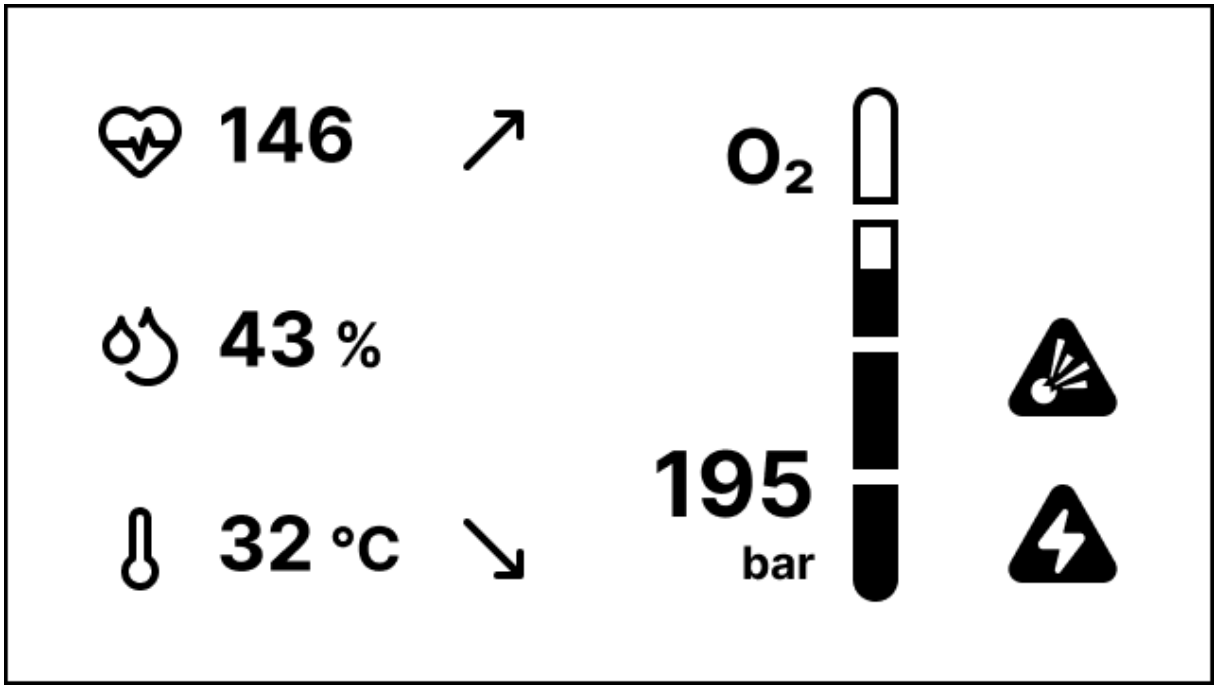

Figure 18: UI concept for a monocular HUD, worn in the upper left of the field of vision.

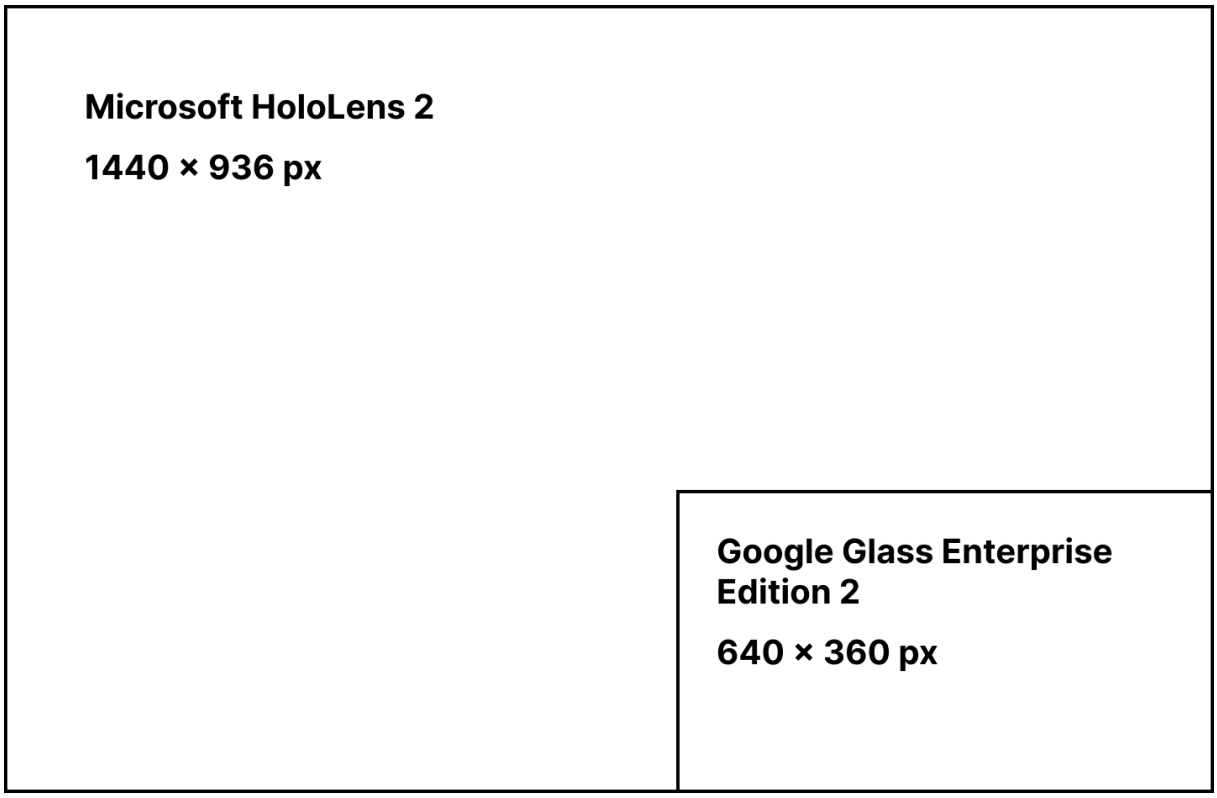

Figure 19: Size comparison of the Microsoft HoloLens 2 and the Google Glass Enterprise Edition 2.

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